NCERT • GRADE 7 • MATHS • PART 2

Connecting the Dots...

C05 • Ganita Prakash Part II

Mean • Median • Range

Dot plots • Double-bar graphs

Textbook problems solved step by step

Textbook solutions first • Visual understanding • Revision

Original student notes based on NCERT. All illustrations are editable LaTeX drawings.

Part A • NCERT questions and worked solutions

Read before you calculate

This chapter has many *statistical investigations*. For open-ended class surveys, graphs drawn from your own measurements, or judgments based on pictures, solutions include an exact **method and a worked example**, not made-up observations. Match the exercise sets to pp. 101, 112, 122–124, and 129–133 in the NCERT textbook.

Set 1 • Figure it Out, page 101

Q1 • Eight attempts at bouncing a ball

Scores: $6, 2, 9, 5, 4, 6, 3, 5$. Sum $= 40$, count $= 8$.
Average $= 40/8 = \boxed{5}$ **bounces.** Check: the average need not be a value appearing in the list, but here it is.

Q2–Q3 • Your bounce attempts and flower survey

These are **data-collection activities** with no fixed numerical answer.
Method: record at least seven scores or daily flower counts; compute mean = sum ÷ number of observations.
Example (not textbook data): $3, 4, 5, 6, 7, 4, 6$ has sum 35, hence mean $= 5$.

Q4 • Nikhil and Sunil's race times

Nikhil: $17, 18, 17, 16, 19, 17, 18$, total 122 seconds. Sunil: $20, 18, 18, 17, 16, 16, 17$, total 122 seconds.
Both average $122/7 = \boxed{17\frac{3}{7}\text{ s}} \approx 17.43$ s. **Neither is faster on average.** Lower time means faster running; here means are tied.

Q5 • Mean school enrolment

Six annual enrolments: $1555, 1670, 1750, 2013, 2040, 2126$.
Total $= 11154$. Mean $= 11154/6 = \boxed{1859}$ students.

Set 2 • Figure it Out, pages 112–113

Q1 • Median onion prices

Yahapur (sorted): $24, 25, 26, 28, 30, 35, 39, 43, 44, 49, 56, 59$. Median $= (35 + 39)/2 = \boxed{37}$.
Wahapur (sorted): $17, 19, 23, 30, 35, 38, 39, 42, 42, 52, 53, 60$. Median $= (38 + 39)/2 = \boxed{38.5}$.
A median compares middle values; the mean is $458/12 \approx 38.17$ for Yahapur, $450/12 = 37.50$ for Wahapur. Different summaries may suggest different comparisons.

Q2 • Domestic animals and pets; absent students

Ignore two missing marks (*not* zero). The 20 recorded numbers have total 72.
Mean $= 72/20 = \boxed{3.6}$; sorted middle pair is $2, 2$, so median $= \boxed{2}$.
The value 25 is a high *outlier*, which pulls the mean upward; median 2 better describes the typical child in this small class survey.

Q3 • Date-palm heights

There are 29 trees. Sum = 1621 ft, so mean = 1621/29 = $\boxed{55.90 \text{ ft (approx.)}}$. The middle (15th) sorted height is $\boxed{56 \text{ ft}}$.
Shorter than the average: count all heights below 55.90: $\boxed{13}$ trees. The range is $67 - 43 = 24$ ft. Draw a dot for every tree at its height from 43 to 67 on a number line.

Q4 • Water-use statistics

Values lie from $\boxed{3.09}$ to $\boxed{20.5}$ L. Their mean = $\boxed{9.71 \text{ L}}$ and median = $\boxed{8 \text{ L}}$.
(a) Neither measure can be between 25 and 30 L because every observation is below 25.
(b) A mean or median cannot be below the minimum or above the maximum of the actual observed dataset.

Q5 • Newborn weights

Boys: 3.5, 4.1, 2.6, 3.2, 3.4, 3.8 kg. Mean = $\boxed{3.43}$ kg, median = $\boxed{3.45}$ kg.
Girls: 4.0, 3.1, 3.4, 3.7, 2.5, 3.4 kg. Mean = $\boxed{3.35}$ kg, median = $\boxed{3.4}$ kg.
Place one dot at each observation; two girl dots sit at 3.4. The data sets overlap considerably: avoid making claims about all babies from these six observations.

Q6 • Grade 5 height dot plots

Read the **mean/median labels** next to the supplied plots, not estimates based only on dot positions.
In the textbook's displayed groups, the measures are:
Whole section: mean 141.21, median 142.5 cm.
Boys: mean 142.05, median 143 cm. Girls: mean 140.14, median 140 cm.
The boys' centre is slightly higher *in this particular sample.* Dot-plot spread and ranges also matter.

Q7 • Sumo wrestlers and dancers

Sumo mean = $\boxed{240.38}$ kg (5 wrestlers). Dancers mean = $\boxed{42.42}$ kg (6 dancers).
Ratio of means = 240.38/42.42 ≈ $\boxed{5.67}$. Thus the mean sumo weight is **about 5.7 times** the mean ballet dancer weight in these lists.

Set 3 • Figure it Out, pages 122–124

Q1 • Animal-speed infographic

(a) Read the horizontal scale (speed in km/h) and tick intervals from the pictured infographic: it is a horizontal-bar comparison, not a time graph.
(b) An example observation: the peregrine falcon's very high diving speed; a further question might compare sustained swimming with brief sprints.
(c) Select two depicted speeds in a near 2:1 ratio, checking the values on the original chart.
(d) A sailfish may be roughly four times as fast as the depicted humpback whale *under the compared speed values*; this chart alone does not prove it is the fastest aquatic animal in the entire world.

This visual question is intentionally qualitative where the infographic does not establish a universal ranking.

Q2 • Grades 5 and 9 superpower preferences

Count each symbol first. The correct frequency table is:

Group	Aerial (a)	Aquatic (w)	Space (s)	None (n)
Grade 5	13	6	2	4
Grade 9	8	6	9	2

Draw paired bars using one unit per response; Grade 5 strongly favours aerial powers, whereas Grade 9 favours space powers.

Q3 • Jodhpur: graph two temperature days

Time	12a	3a	6a	9a	12p	3p	6p	9p
Day 1	20	18	16	20	26	34	30	24
Day 2	37	34	30	33	37	43	42	39

Use the stated **scale 1 unit = 4°C**, paired bars for each time. Day 1 is consistent with a milder season (e.g. winter/early spring), Day 2 with a hot summer day; *the exact month cannot be proved solely from these temperatures.*

Q4 • Electric-vehicle registrations

(a) Draw Gujarat's three bars at 69,000, 89,000, 78,000 and Delhi's at 62,000, 74,000, 81,000 (2022, 2023, 2024).
(b–e) Compare the three years for each state. For other states, read the approximate heights in NCERT Fig. p. 124, keeping the stated 25,000 guideline increments; the bars are approximate rather than exact counts.
(f) No: a small *change* between bars does not imply few registrations in each year; bar *height* gives the annual total, and the difference gives the change.

Set 4 • Figure it Out, pages 129–133

Q1 • Pocket-count dot plots

Read the dots: boys' pocket counts are 3 (once), 4 (four), 5 (five), 6 (twice); girls' counts are 0 (once), 2 (once), 3 (four), 4 (five), 5 (once), 6 (once).
(a) False: girls' range $6-0=6$ is larger than boys' range $6-3=3$.
(b) True: boys' median $=5$, girls' median $=4$.
(c) False: girls' mean $=45/13\approx 3.46$ is less than boys' mean $=56/12\approx 4.67$.
(d) False: both maximum counts are 6 pockets.
Only statement (b) is true.

Q2 • Four games: three players

A: $14+16+10+10=50$, mean $=\boxed{12.5}$. B: $0+8+6+4=18$, mean $=\boxed{4.5}$. C: $8+11+13=32$ in *three* games, mean $=\boxed{10.67}$.
A has the highest mean, so A is strongest by average points per game. A zero score counts as a game played; *did not play* does not.

Q3 • Two quiz-score groups

Group 1: mean $=\boxed{73}$, median $=\boxed{78.5}$. Group 2: mean $=\boxed{75.67}$, median $=\boxed{79}$. Group 2 has the slightly higher mean and median; group 1 has the lower minimum (39 vs 47).

Q4 • Sports: watching vs participating

Plot five pairs of bars in category order (cricket, basketball, swimming, hockey, athletics): watching $1240, 470, 510, 430, 250$; participating $620, 320, 320, 250, 105$.
Choose a legible scale, e.g. 1 tick = 200 students, and a separate legend. **Cricket** has the highest count for both. More students watch than participate in each category.

Q5 • Divide 17 students by height

Ordered heights have middle (9th) value $\boxed{115\text{ cm}}$. The median is 115 cm, but **five children share 115 cm**, so it is impossible to split all students into two groups with exactly 8 below and 8 above 115 cm. Even the total of 17 is odd, so exact equal groups require leaving one person out or special handling of ties.

Q6 • Your class heights

No fixed answer; record everyone's measured height, sort the list, calculate mean and median, and build a dot plot. Do not assume one class matches another.

Q7 • Other Grade 7 section's mean

Option (d): it cannot be determined. Having 15 boys and 15 girls in both classes does not force their average heights to be the same.

Q8 • Skyscrapers in the pictured chart

(a) Approximate from its bars: New York about **310**, Tokyo about **165**, London about **30** buildings above 150 m.
(b)(i) The graphic shows 12 listed cities above Mumbai; that is true *within the displayed rankings*, not a proof about every world city.
(ii) Seven listed cities are below Mumbai in this display.
(iii) False inference: the chart counts buildings above 150 m; it does not identify the location of the world's single tallest building.

Q9 • Estimate then measure five objects

This is a **hands-on task**. Estimate and measure the pen, eraser, palm, geometry box and notebook. For each object find $|\text{estimate} - \text{measured}|$; the mean absolute difference is the sum of five differences divided by 5. Draw paired bars of your measurements; there is no universal answer.

Q10 • Aditi's Sudoku times

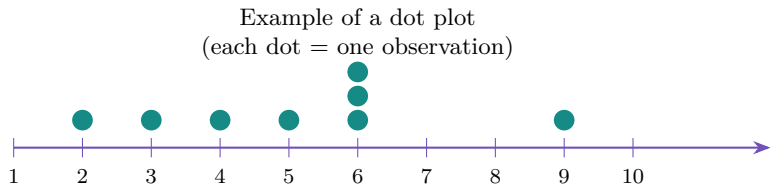
Week 1 (9 times): mean = $\boxed{360\text{ s}}$, median = $\boxed{360\text{ s}}$.
Week 2 (8 times): mean = $\boxed{278.75\text{ s}}$, median = $\boxed{275\text{ s}}$.
Draw separate dots for each week; Week 2 generally shows faster solutions (lower times), though one time of 380 s is relatively high.

Q11–Q13 • Projects: sentences, names and time

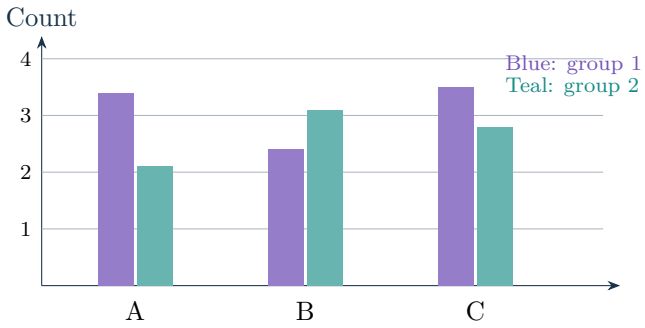
These are **student data-collection projects**, so no unique numerical results exist.
Q11: Compare two textbook pages' sentence lengths, or classmate-name lengths and first letters; make plots and calculate mean/median.
Q12: Record daily outings for a month; inspect range, typical value and dot plot.
Q13: Collect family heights or estimates of 1 and 3 minutes; make paired graphs, compare family means and note outliers. Use your actual observations.

Part B • Visual understanding

1. Mean, median and range are different.



2. Compare two categories using paired bars.



The three checks for any graph

Axes: what is shown and in which units? **Scale:** equal spaces must mean equal numerical steps. **Claim:** does the data really justify it, or is it only an estimate? A graph of a sample does not prove a claim about every person.

Part C • Key methods and common errors

Formula card

$$\text{Mean} = \frac{\text{sum of observed values}}{\text{number of observed values}}$$

$$\text{Range} = \text{maximum} - \text{minimum}$$

Median: first sort; odd count → middle value; even count → average the two middle values.

- **Missing entry is not zero.** Exclude "did not play" from both total and count.
- **Mean is not always one of the data values.** It may be decimal or fraction.
- **Outliers can move the mean.** Median often stays closer to the centre of most values.
- **Never say one variable causes another** just because the graph changes together.
- **Graph title, axes, scale and legend** are needed for a readable visualisation.

Quick self-check: Mean of $2, 4, 6 = 4$; median of $\frac{1}{5}, 2, 10 = 2$; median of $1, 3, 8, 9 = 5.5$. Outliers often shift the **mean**; each plotted dot represents one observation.